How to Print PA6-CF (Nylon 6 Carbon Fiber) Filament

- Printer Requirements:
  - Use a hardened steel or ruby-tipped nozzle to withstand carbon fiber abrasiveness.
  - An enclosed printer is recommended to prevent warping.
  - Use a direct drive extruder for better control, though high-quality Bowden setups can also work.
- Recommended Print Settings:
  - Nozzle Temperature: 260–290°C (Check manufacturer specs)
  - Bed Temperature: 70–110°C
  - Layer Height: 0.4 mm or greater (due to fiber size)
- Print Surface & Adhesion:
  - Use PEI.
  - Apply a glue stick or a PA-specific adhesive to prevent sticking too hard or warping.
- Drying the Filament:
  - Critical step – Nylon is extremely hygroscopic.
  - Dry filament at 80-90°C for 12 hours before printing.
  - Check for <20% relative humidity before printing.
  - Store in a dry box or vacuum-sealed bag with desiccant.
- Speed & Retraction:
  - Print Speed: 60 mm/s (slower for better quality)
  - Retraction: 1–2 mm @ 20–40 mm/s (tune for your setup to avoid stringing)
- Notes
  - The body shrinks enough to warrant printing at 1.01-1.02 scale.
  - For a looser battery cage fit, scale the print 0.98 - 0.99.

How to Print TPU (Thermoplastic Polyurethane) Filament

- Printer Requirements:
  - Direct drive extruder is highly recommended – TPU is flexible and can jam in Bowden tubes. 95A should be fine.
  - Use a printer with a fully constrained filament path to prevent buckling.
  - Works well on open-frame printers – no enclosure required.
- Recommended Print Settings:
  - Nozzle Temperature: 210–240°C (check your specific TPU brand)
  - Bed Temperature: 30–60°C (optional; helps first-layer adhesion)
  - Layer Height: 0.1–0.3 mm (0.2 mm is typical)
- Print Surface & Adhesion:
  - Sticks well to PEI.
  - Use a brim or skirt to help with initial adhesion and oozing control.
- Drying the Filament:
  - TPU is hygroscopic; it absorbs moisture easily.
  - Dry at 40–50°C for 12 hours before printing.
  - Check for <20% relative humidity before printing.
  - Store in a dry box or airtight container with desiccant.
- Speed & Retraction:
  - Print Speed: 15–60 mm/s (go slow to avoid issues)
  - Retraction: Minimal (0.5–2 mm @ 10–25 mm/s); too much can cause clogs or jams
  - Enable avoid crossing walls.
  - Disable coasting, combing, and z-hop for more consistent extrusion

How to Remove Supports from 3D Prints

- Let the Print Cool First:
    - Allow the print to cool fully before handling—supports are easier to break off cleanly when the material is rigid.
- Start with Your Hands:
    - Gently break away large support structures by hand.
    - Flex the part slightly (if it's not brittle) to loosen supports from internal cavities.
    - Be sure to support small features when removing supports.
- Use the Right Tools:
    - Flush cutters or diagonal pliers for precise nips
    - Deburring tool or X-Acto knife for edge cleanup
    - Needle-nose pliers for tight or internal areas
    - Tweezers for stringy supports in fine details
- Use Support Interface Layers (Slicer Tip):
    - When slicing, enable support interface layers (denser contact layers) for easier, cleaner breakaway.
    - Consider "air gap" settings for easier removal—typically 0.2 mm for PLA, more for PETG or TPU.
- Print Orientation Matters:
    - Orient parts to minimize supports or make them easier to remove (e.g., flat surfaces facing down).
- Stay Safe:
    - Wear gloves and eye protection when using sharp tools.
    - Cut away from your body—many support injuries come from rushed removal.

How to Wire the Stepper Motors and an Encoder

- Wire Lengths:
    - Switch: >6inches
    - Steering Encoder: >4 inches
    - Steering Motor; >4 inches
    - Left Motor: > 5 inches
    - Right Motor: >5 inches
- Wire Each Stepper Motor:
    - Use stepper motor datasheets or a multimeter to identify coil pairs.
      Connect each motor's A+ A– and B+ B– wires to the correct driver terminals.
      Keep wire lengths manageable and avoid crossing wires when possible.
- Connect the Encoders:
    - Use the 3-pin encoder for steering and the 5-pin encoders coming out of the stepper for the drive steppers.
    - When soldering the steering encoder there are 2 ways to make the encoder work correctly.

      1. Solder a wire between VCC and GP0.

      OR

      2. Remove resistor 4.
- Secure Your Wiring:
    - Make sure the steering encoder is attached to the bottom cover and routed through the vehicle before attaching the bottom cover. Label both ends of each wire for easy debugging later.
- Stay Safe:
    - Disconnect power before making any changes.
    - Avoid touching stepper drivers when powered—they get hot quickly.
    - Keep loose wires away from fan blades or moving parts.

Reminders for Assembling the Car

- Check All Parts First:
  - Lay out every printed piece and compare against your assembly plan or parts list.
  - Verify that parts are cleanly printed, supports are removed, and undamaged before starting.
- Check tolerances:
  - Lightly sand mating surfaces if parts are too tight or warped.
- Use Adhesives:
  - Superglue (CA glue) for plastic interfaces.
  - Loctite 425 for metal-to-metal interfaces.
  - Apply sparingly — a little goes a long way and avoids mess.
- Check Alignment and Motion:
  - Spin wheels to ensure smooth rotation and even clearance.
  - Test steering movement before locking everything in place.
  - Steering should tighten until the shoulder bolt almost peeks out from the nylon insert in the Nylock nut.
- Finish and Detail (Optional):
  - Sand seams, then prime and paint if desired.
  - Apply decals or stickers after clear-coating for best results.
- Stay Safe:
  - Wear eye protection and gloves when cutting, sanding, or using adhesives.
  - Work in a well-ventilated area when using glue or paint fumes.

3DP Parts List

3D Prints, Recomended Materials, and Quantity		
Parts	Filament	Qty
Main Body	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Bottom Cover	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Front Bumper	Bambu 95A TPU, Markforged TPU	1
Rear Bumper	Bambu 95A TPU, Markforged TPU	1
Sensor Plate	Bambu PETG-CF, Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Camera Mount	Bambu PETG-CF, Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Wheel Hub	Bambu PETG-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	2
Wheel Hub Stub	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	2
Wheel Hub Washer	Bambu PETG-CF, Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	4
Steering Rod	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Motor Connector 1	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Motor Connector 2	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
12t Pinion	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
55t Gear	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Pinion Clamp	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Steering dead axle	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Stepper Spacer	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1
Front Wheel Rim	Bambu PETG-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	2
Rear Wheel	Bambu PETG-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	2
Tire	Bambu 95A TPU, Markforged TPU	4
Front Right Shroud	Bambu PETG, Polymaker HT PLA	1
Front Left Shroud	Bambu PETG, Polymaker HT PLA	1
Rear Right Shroud	Bambu PETG, Polymaker HT PLA	1
Rear Left Shroud	Bambu PETG, Polymaker HT PLA	1
Tire	Bambu 95A TPU, Markforged TPU	4
Lidar Clip	Bambu PA6-CF, Markforged Onyx, 3DXTech Obsidian, Polymaker HT PLA	1